

MERIDIAN FAMILY OFFICE · COPILOT v10

AI Copilot — Test Suite & Results

Verifying the deterministic engine, the deliverable layer, and the end-to-end app

135/135 tests passed · 16 July 2026

Runs fully offline in DEMO_MODE — no API key, no network, no billing.

Coverage at a glance

Central invariant: every number is deterministic; documents and instructions shape prose, never figures.

Engine	Deliverable	App	Business
3 files · 30 tests Parsing · reconciliation · suitability · grounded Q&A	3 files · 23 tests Doc ingestion · portable prompt · narrative · deck render	1 file · 19 tests Streamlit end-to-end across every view	2 files · 63 tests Client macro views: rates · war · tax → prose, not the numbers

What's proven end-to-end

- The v10 fix: research/other documents and tactical instructions change the deterministic commentary and every exported deck (HTML, PPTX, PDF).
- The same book reads differently by mandate: advisory flags, discretionary hard-blocks.
- Reconciliation catches the ~\$436 custodian break instead of trusting it silently.
- A macro view as tactical text / documents reshapes the commentary but never moves the figures; a selected macro OVERLAY deliberately shifts the recommendation via a bounded, provenanced target tilt (rates → cash into bonds; risk-off → gold & cash up, equity down).

Statement ingestion

9/9 PASS

Engine · tests/test_statement_parser.py

Parse the three custodian formats, reconcile against each statement's own total, convert FX to USD, and keep a PII-free continuity key. Every downstream number rides here.

- ✓ Parse custodian A (USD, per-share CSV) into valued positions.
- ✓ Semicolon CSV, European decimals, EUR/CHF → converted to a USD base.
- ✓ JSON custodian carries illiquid alternatives + a Lombard loan (a liability).
- ✓ One sample deliberately fails its checksum by ~\$436 — it must be flagged, not silently trusted.
- ✓ The ~\$436 break on custodian B reconciles under a generous tolerance but fails under a tight one — proving the checksum knob actually gates the flag.
- ✓ Continuity key is deterministic across re-parses and carries no PII.
- ✓ Continuity check compares month-to-month keys.
- ✓ Consolidate all three custodians into one holdings list.
- ✓ gross – debt = net, and gross is the sum of position base values.

The deterministic policy checks. The mandate decides what a breach MEANS; plus concentration, liquidity, unhedged-FX, hard regulatory blocks and instruction-diff.

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| ✓ A single overweight equity book, evaluated under each mandate, yields the mandate-appropriate enforcement on the band breach. | ✓ Sanctioned instrument hard blocks even execution only. |
| ✓ A balanced book inside every band produces no actionable breach. | ✓ Complex product appropriateness blocks when not allowed. |
| ✓ A single stock above the 10% single-security cap is caught; a diversified fund of the same weight is not. | ✓ Client asked for a treasury sleeve; the proposal holds a corporate-credit bond fund of the wrong duration → a deviation flag. |
| ✓ Parked liquidity (money-market) is exempt from the single-position cap. | ✓ Instruction diff satisfied when matched. |
| ✓ Below the min-liquid floor (here mostly illiquid alternatives) is flagged. | ✓ Worst enforcement ranking. |
| ✓ Economic FX exposure above the max-unhedged cap is flagged. | ✓ The three-custodian sample book, under its advisory profile, lands on a 'flag' gate (bands breached, nothing hard-blocked). |

Routing picks the right deterministic tool from the phrasing; answers stay grounded in computed facts; the router never crashes on odd input.



'Net worth' routes to the totals tool.



An FX / currency question routes to the currency tool.



A suitability / mandate question routes to the suitability tool.



A 'largest position' question routes to the concentration tool.



A liquidity question routes to the liquidity tool.



A reconciliation question routes to the data-quality tool.



Every answer carries facts tied to the computed book.



The deterministic briefing fact-sheet spans every domain.



The router must always resolve to SOME tool — never crash on odd input.

The research / other uploaders read text deterministically so content can be folded into the prompt as advisory context — never a source of figures.

- ✓ Plain text and Markdown read cleanly.
- ✓ HTML is stripped to readable text (tags removed).
- ✓ An .eml email body is extracted.
- ✓ CSV and JSON pass through as text.
- ✓ An unsupported file type fails gracefully (no crash).
- ✓ Very large documents are truncated (MAX_CHARS) and marked, never dropped.
- ✓ The extraction result always has the same shape.

The portable prompt bundles every input; the keyless commentary REFLECTS the documents (the v10 fix) while only ever quoting the computed numbers.

✓ The prompt carries every block (FACTS, intake, holdings, docs, tactical).	✓ The commentary quotes short excerpts of the tactical text and docs.
✓ Document text is embedded verbatim in the prompt.	✓ The keyless summary must use the model's own numbers verbatim and never invent a figure that isn't already somewhere in the computed model.
✓ Without documents, that content is absent from the prompt.	✓ The commentary carries the 'not investment advice' disclaimer.
✓ The FACTS block is valid JSON and holds the numbers.	✓ The prompt that generates the commentary slide must NOT ask the model to build a deck / produce files — otherwise the model returns the slide-by-slide spec, which leaks onto the commentary slide.
✓ v10 fix: documents change the deterministic commentary.	✓ The portable whole-deck prompt (a separate artifact) still asks for the full deck — that behaviour is intentional and must stay distinct from the commentary.

Proposal deck render

6/6 PASS

Deliverable · tests/test_generate_proposal.py

HTML / PPTX / PDF renderers agree, figures are copied from the model, and documents change every exported format.

- ✓ All three renderers (HTML/PPTX/PDF) produce valid output.
- ✓ The deck shows the computed net-worth & gross figures.
- ✓ The commentary slide appears only when a narrative is set.
- ✓ v10 fix: documents change the HTML, PPTX and PDF exports.
- ✓ With documents, the deck surfaces the client's liquidity event.
- ✓ The deck carries the 'not independently verified' disclaimer.

Drive the real Streamlit app across the views a banker walks through: load samples, review each view, change the mandate & tactical instructions — and never error.

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| ✓ The app boots with no book loaded. | ✓ Selecting a macro overlay tilts the target the rebalance chases, and the Proposal view says so — the up-sleeve's target rises and the down-sleeve's falls. (2×) |
| ✓ Each sample statement loads into the Overview view. (3×) | ✓ The Proposal view shows the assembled prompt and commentary. |
| ✓ Overview / Holdings / Suitability / Proposal all render. (4×) | ✓ Tactical instructions flow into the on-screen prompt. |
| ✓ Proposal does not auto generate commentary. | ✓ Under discretionary, band breaches HARD-BLOCK; under advisory they only flag. The same book therefore reads differently on the Suitability view. (2×) |
| ✓ Generate button produces commentary on click. | ✓ A single-custodian book still produces a proposal. |
| ✓ No overlay → no tilt banner on the proposal. | ✓ The Proposal view with no book degrades gracefully. |

What happens to the output on a client macro view — a rate hike/cut, a war (or its end) prompting alternatives, a tax rise/cut: it reaches the AI, shapes the commentary, and never moves the grounded figures.

- ✓ A client macro view (rate/war/tax) reaches the AI verbatim. (12×)
- ✓ The macro view shapes the advisory commentary. (12×)
- ✓ A macro opinion shapes prose, never figures: the FACTS block (every number the deck shows) is byte-for-byte identical with and without the scenario. (12×)
- ✓ 'What if rates are hiked?' → the deterministic tool states the fixed-income exposure but refuses to estimate a +100bp impact it cannot source.
- ✓ A rate-move question routes to the rate tool.
- ✓ The scenario moves the deck's prose, not its tables.

A client macro view MOVES the recommendation: each overlay is a fixed, bounded, provenanced tilt of the target allocation — the rebalancer chases it, so trades shift in the scenario's direction while every figure stays deterministic.

- ✓ Each overlay moves the target sleeves in the documented direction. (5×)
- ✓ A tilt sums to zero — the book stays fully invested. (5×)
- ✓ No sleeve is driven negative by a tilt. (5×)
- ✓ 'No overlay' changes nothing.
- ✓ An unknown overlay falls back to a no-op.
- ✓ Reported changes name only moved sleeves; the summary is provenanced. (5×)
- ✓ Tilts are modest: no sleeve moves more than 8pp from its strategic target, so a tilt never breaches a normal $\pm 15\%$ suitability band on its own.
- ✓ If a tilt would drive a sleeve negative, it clamps at 0 and renormalises so the total is preserved (no phantom allocation).

How to run

python3 -m pytest

python3 -m pytest tests/test_suitability.py -v

135/135 passed · offline, deterministic, repeatable